Practice Set A

Class 12 Mathematics, Chapter 1: Relations and Functions

Maximum marks: **52** **20** questions

Name _____ Date _____ Score _____ / 52

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, CBSE 2024 (65/1/3) Q7 — 1 mark

A relation R defined on a set of human beings as $R = \{(x, y) : x \text{ is } 5 \text{ cm shorter than } y\}$ is:

(A) reflexive only

(B) reflexive and transitive

(C) symmetric and transitive

(D) neither transitive, nor symmetric, nor reflexive

2 MCQ, CBSE 2024 (65/1/2) Q13 — 1 mark

A relation R defined on set $A = \{x : x \in \mathbb{Z} \text{ and } 0 \leq x \leq 10\}$ as $R = \{(x, y) : x = y\}$ is given to be an equivalence relation. The number of equivalence classes is:

(A) 1 **(B)** 2

(C) 10 **(D)** 11

3 MCQ, CBSE 2024 (65/3/1) Q2 — 1 mark

Let $\mathbb{R}_+$ denote the set of all non-negative real numbers. Then the function $f : \mathbb{R}_+ \to \mathbb{R}_+$ defined as $f(x) = x^2 + 1$ is:

(A) one-one but not onto **(B)** onto but not one-one

(C) both one-one and onto **(D)** neither one-one nor onto

4 MCQ, CBSE 2024 (65/1/1) Q1 — 1 mark

A function $f : \mathbb{R}_+ \to \mathbb{R}$ (where $\mathbb{R}_+$ is the set of all non-negative real numbers) defined by $f(x) = 4x + 3$ is:

(A) both one-one and onto **(B)** onto but not one-one

(C) one-one but not onto **(D)** neither one-one nor onto

5 MCQ, Board-pattern — 1 mark

Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b\}$. The number of onto functions from A to B is:

(A) 16 **(B)** 14

(C) 12 **(D)** 8

Q No.	Question	Marks
6	Assertion-Reason, CBSE 2024 (65/4/1) Q19 **Assertion (A):** The relation $R = \{(x, y) : (x + y) \text{ is a prime number and } x, y \in \mathbb{N}\}$ is not a reflexive relation. **Reason (R):** The number $2n$ is composite for all natural numbers n. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, CBSE 2024 (65/1/1) Q26(b) A function f is defined from $\mathbb{R} \to \mathbb{R}$ as $f(x) = ax + b$, such that $f(1) = 1$ and $f(2) = 3$. Find function $f(x)$. Hence, check whether function $f(x)$ is one-one and onto or not.	2 marks

Level 2

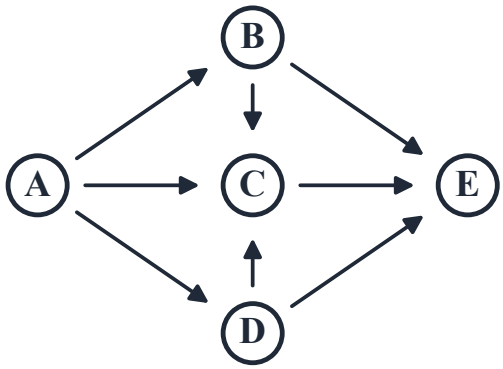
19 marks

8	Short answer, PYQ-style Show that the relation R on the set $\mathbb{Z}$ of integers, given by $R = \{(a, b) : 2a + 3b \text{ is divisible by } 5\}$, is an equivalence relation.	2 marks
9	Short answer, CBSE 2023 (65/1/1) Q32 A relation R is defined on the set of real numbers $\mathbb{R}$ as $R = \{(x, y) : x \cdot y \text{ is an irrational number}\}$. Check whether R is reflexive, symmetric and transitive or not.	3 marks
10	Short answer, Board-pattern Let $A = \{1, 2, 3, \ldots, 9\}$ and R be the relation in $A \times A$ defined by $(a, b)\, R\, (c, d)$ if $a + b = c + d$ for $(a, b), (c, d) \in A \times A$. Prove that R is an equivalence relation. Also obtain the equivalence class $[(3, 5)]$.	3 marks
11	Short answer, PYQ-style Let $f : \mathbb{N} \to \mathbb{N}$ be defined by $f(n) = \begin{cases} n + 1, & \text{if } n \text{ is odd} \\ n - 1, & \text{if } n \text{ is even} \end{cases}$. Show that f is both one-one and onto.	3 marks
12	Short answer, CBSE 2024 (65/2/3) Q33(a) Show that a function $f : \mathbb{R} \to \mathbb{R}$ defined as $f(x) = x^2 + x + 1$ is neither one-one nor onto. Also, find all the values of x for which $f(x) = 3$.	3 marks
13	Short answer, CBSE 2023 (65/4/1) Q34(b) Let $f : \mathbb{R} - \left\{-\frac{4}{3}\right\} \to \mathbb{R}$ be a function defined as $f(x) = \frac{4x}{3x + 4}$. Show that f is a one-one function. Also, check whether f is an onto function or not.	3 marks
14	Short answer, CBSE 2024 (65/1/1) Q26(a) A relation R on set $A = \{1, 2, 3, 4, 5\}$ is defined as $R = \{(x, y) : \lvert x^2 - y^2 \rvert < 8\}$. Check whether the relation R is reflexive, symmetric and transitive.	2 marks

Q No.	Question	Marks
Level 3		**25 marks**
15	Short answer, NCERT Ex 1.1 Q12 Show that the relation R defined in the set A of all triangles as $R = \{(T_1, T_2) : T_1 \text{ is similar to } T_2\}$ is an equivalence relation. Consider three right angle triangles T_1 with sides $3, 4, 5$, T_2 with sides $5, 12, 13$ and T_3 with sides $6, 8, 10$. Which triangles among T_1, T_2 and T_3 are related?	3 marks
16	Long answer, CBSE 2023 (65/4/1) Q34(a) If $\mathbb{N}$ denotes the set of all natural numbers and R is the relation on $\mathbb{N} \times \mathbb{N}$ defined by $(a, b)\, R\, (c, d)$ if $ad(b + c) = bc(a + d)$, show that R is an equivalence relation.	5 marks
17	Long answer, Board-pattern Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x\|x\|$ is both one-one and onto.	5 marks
18	Long answer, CBSE 2024 (65/3/1) Q32 A relation R on set $A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ is defined as $R = \{(x, y) : x + y \text{ is an integer divisible by } 2\}$. Show that R is an equivalence relation. Also, write the equivalence class $[2]$.	5 marks
19	Short answer, CBSE 2023 (65/2/2) Q34 A function $f : [-4, 4] \to [0, 4]$ is given by $f(x) = \sqrt{16 - x^2}$. Show that f is an onto function but not a one-one function. Further, find all possible values of a for which $f(a) = \sqrt{7}$.	3 marks

Q No.	Question	Marks
20	Case study, CBSE SQP 2025-26 Q36	4 marks

Case Study: A city's traffic management department is planning to optimize traffic flow by analyzing the connectivity between various traffic signals. The city has five major spots labelled A, B, C, D and E. The department has collected the following data regarding one-way traffic flow between spots: traffic flows from A to B, A to C and A to D; from B to C and B to E; from C to E; from D to E and D to C. The department wants to represent and analyze this data using relations and functions, taking $(X, Y) \in R$ when traffic flows from X to Y, on the set $S = \{A, B, C, D, E\}$.



arrow $X \to Y$: one-way traffic from X to Y

(i) Is the traffic flow reflexive? Justify. 1 mark

(ii) Is the traffic flow transitive? Justify. 1 mark

(iii)(a) Represent the relation describing the traffic flow as a set of ordered pairs. Also state the domain and range of the relation. 2 marks

OR

(iii)(b) Does the traffic flow represent a function? Justify your answer. 2 marks

Answer key

Chapter 1: Relations and Functions, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(D) neither transitive, nor symmetric, nor reflexive
2	(D) 11
3	(A) one-one but not onto
4	(C) one-one but not onto
5	(B) 14
6	(C) A is true, but R is false
7	$f(x) = 2x - 1$; it is both one-one and onto.

Level 2

8	R is reflexive, symmetric and transitive, hence an equivalence relation.
9	R is symmetric, but neither reflexive nor transitive.
10	R is an equivalence relation; $[(3,5)] = \{(1,7),(2,6),(3,5),(4,4),(5,3),(6,2),(7,1)\}$
11	f is both one-one and onto (bijective).
12	Neither one-one nor onto; $f(x) = 3$ for $x = 1$ and $x = -2$.
13	f is one-one but not onto ($\frac{4}{3}$ has no pre-image).
14	R is reflexive and symmetric, but not transitive.

Level 3

15	R is an equivalence relation; only T_1 and T_3 are related.
16	R is reflexive, symmetric and transitive, hence an equivalence relation.
17	f is both one-one and onto (bijective).
18	R is an equivalence relation; $[2] = \{-4,-2,0,2,4\}$
19	f is onto but not one-one; $a = 3$ or $a = -3$.
20	(i) Not reflexive (ii) Not transitive (iii)(a) 8 ordered pairs; domain $\{A,B,C,D\}$, range $\{B,C,D,E\}$ (iii)(b) Not a function